

Donghee Hong

Suwon-si, Republic of Korea

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Research Interests

Visual analytics, human-computer interaction, user-centered design, interactive visualization, interactive ML

Education

University of Maryland, College Park (UMD)

PH.D. IN COMPUTER SCIENCE (STARTING AUG 2026)

College Park, MD, USA

Incoming

Sungkyunkwan University (SKKU)

M.S. IN ARTIFICIAL INTELLIGENCE

Suwon, Korea

Mar 2024 - Feb 2026

- Cumulative GPA: 4.5/4.5

Sungkyunkwan University

B.S. IN COMPUTER SCIENCE AND ENGINEERING

Suwon, Korea

Mar 2020 - Feb 2024

- Cumulative GPA: 4.35/4.5 (98.5%), Major GPA: 4.38/4.5
- Graduated Summa Cum Laude, Class Rank: 2/160

Research Experience

Human-Data Interaction Research Group, UMD

RESEARCH ASSISTANT (ADVISER: PROF. LEO ZHICHENG LIU)

College Park, USA

July 2026 - Present

Interactive Data Computing Laboratory (IDCLab), SKKU

RESEARCH ASSISTANT (ADVISER: PROF. JAEMIN JO)

Suwon, Korea

Aug 2022 - June 2026

Data-driven AI Security HCI (DASH) Laboratory, SKKU

RESEARCH ASSISTANT (ADVISER: PROF. SIMON SUNG-IL WOO)

Suwon, Korea

Dec 2020 - Jun 2022

Publications

UNDER REVIEW

- [7] **Hong, D.**, Kim, M., Cha, S., & Jo, J. (IEEE VIS 2026 Short Papers). TuneScape: Visualizing Multi-Tuner Parameter Optimization Logs.

PUBLISHED

- [6] **Hong, D.**, Kim, M., Cha, S., & Jo, J. (2026). Symetra: Visual Analytics for the Parameter Tuning Process of Symbolic Execution Engines. *EuroVis 2026*, accepted.
- [5] **Hong, D.**, Lee, B., Seo, J., & Jo, J. (2026). HyPockeTuner: Bringing Hyperparameter Optimization to Mobile Devices. *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*, 1–18.
- [4] **Hong, D.**, Kim, M., Cha, S., & Jo, J. (2025). Symetra: Visual Analytics for Tuning Symbolic Execution Engines. *IEEE VIS (Poster)*.
- [3] Park, S., Park, J., & **Hong, D.** (2024). LLM-Based Text-to-Image Prompt Conversion System for Custom Wheelchair Spoke Guard Design. *Proceedings of the Korean Computer Science Society Conference*, 918–920.
- [2] Lee, S., Ko, D., Park, J., Shin, S., **Hong, D.**, & Woo, S. S. (2022, May). Deepfake Detection for Fake Images with Facemasks. *Proceedings of the 1st Workshop on Security Implications of Deepfakes and Cheapfakes*, 27–30.
- [1] An, J., Kim, J., Lee, H., Kim, J., Kang, J., Shin, S., Kim, M., **Hong, D.**, & Woo, S. S. (2021, December). VFP290k: A Large-Scale Benchmark Dataset for Vision-Based Fallen Person Detection. *The Thirty-Fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2)*.

Presentation

Hong, D. (2026, Apr). HyPockeTuner: Bringing Hyperparameter Optimization to Mobile Devices. ACM CHI Conference on Human Factors in Computing Systems (CHI 2026), Barcelona, Spain.

Hong, D. (2024, Oct). ViSTrics: A visual analytics tool for software testing. Korean Visualization Workshop (K-VIS), Seoul, Korea.

Teaching Experience

TEACHING ASSISTANT

Introduction to Human-Computer Interaction (Fall 2024, Fall 2025); Information Visualization (Spring 2025); Problem Solving (Fall 2025); Computer Programming for Engineers (Spring 2023, Spring 2025); Python Programming Orientation for Freshmen (2021-2023)

TUTORING AND MENTORING

SKKU Peer Tutoring Program (Spring 2021); SKKU Software Mentoring Program (2020-2021)

Awards and Honors

SCHOLARSHIPS

Dean's Fellowship , UMD	<i>Sep 2026 - Aug 2028</i>
Annual fellowship of \$2,500 for two years	
Graduate Scholarship , SBS CULTURAL FOUNDATION	<i>Sep 2024 - Feb 2026</i>
- Full tuition and annual stipend of \$8,225 - Around 15 students selected nationwide (acceptance rate: 4%)	
Graduate Scholarship , DAESANG FOUNDATION	<i>Mar 2024 - Jun 2024</i>
Stipend of \$4,800 per semester	
The Presidential Science Scholarship , KOREA STUDENT AID FOUNDATION	<i>Mar 2022 - Dec 2023</i>
- Full tuition and annual stipend of \$3,500 - Approximately 150 students selected nationwide	
SKKU Software Scholarship , SKKU	<i>Mar 2020 - Dec 2022</i>
Full tuition for two years (contingent on maintaining an overall GPA above 3.5)	
Additional Internal Scholarships , SKKU	
Extracurricular Challenge Scholarship (2023); Student Success-Leadership Scholarship (2021, 2022); Department Recommendation Scholarship (Fall 2021)	

AWARDS

Honorable Mention, Graduate Student Paper Award , SKKU	<i>Dec 2025</i>
Grand Prize, Co-Deep Learning Project Competition , SKKU	<i>Dec 2023</i>
Ranked top 2 out of 16 teams	
Honorable Mention, SW-Centered College AI Competition , SW-CENTERED UNIVERSITY ASSOCIATION	<i>Dec 2023</i>
Ranked top 8 out of 194 teams	
Dean's List Award, College of Computing and Informatics , SKKU	<i>Spring & Fall 2020, Fall 2021</i>

Leadership and Extracurricular Activities

SKKU-BA-DIVE Program

TEAM LEADER

- Led field research and interviews with industry professionals in IT and AI

London, UK
Jan 2023

Uni-DTHON Competition

DATA-THON TRACK LEADER

Seoul, Korea
Dec 2021 - Dec 2022

- Coordinated AI competition planning across six major universities in South Korea, designing competition problems and implementing leaderboard evaluation

Student Council, College of Computing and Informatics

VICE PRESIDENT

Suwon, Korea
Nov 2021 - Nov 2022

- Organized in-person academic events and facilitated communication between students, faculty, and partner universities

University Student Council Joint Committee

COLLEGE REPRESENTATIVE

Seoul, Korea
Nov 2021 - Nov 2022

- Represented the College of Computing and Informatics to discuss and decide on university development issues

Global Challenge Program

DEPUTY TEAM LEADER

Silicon Valley, USA
Jun 2022

- Conducted field research and interviews with IT professionals

Community Service

VOLUNTEER

Suwon, Korea
2020 -2024

- Over 100 hours of community service

Technical Skills

Research Methods	Semi-structured interviews, Case Studies, Controlled Lab Studies, Formative Studies, Thematic Analysis
Programming	TypeScript and JavaScript, Python, Java, C and C++
Visualization & Frontend	D3.js & visx, React, Tailwind CSS, HTML/CSS
Data Analysis & Backend	Pandas, PyTorch, NumPy, Scikit-learn, FastAPI
Tools	Git, Figma, LaTeX, Adobe Premiere Pro